How Many Sensors Are Enough? An Exhaustive Sensor-Subset Analysis for Rotating Machinery Fault Diagnosis

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***Abstract*—Condition-monitoring systems are usually designed by adding sensors until diagnosis is good enough, and the cost of that choice is rarely measured. We ask the opposite question: how far can instrumentation be reduced before diagnosis degrades? On the MaFaulDa dataset we evaluate *all* 127 non-empty subsets of seven sensors with one fixed Random Forest, under cross-validation holding out whole rotation-speed bands, with every subset chosen from training data alone and three hypotheses fixed in advance. Macro-F1 falls from 0.9917 ± 0.0066 with seven sensors to 0.8799 for the best single sensor, and removing a sensor never helped by more than the 0.01 tolerance. Four sensors reach 0.984 ± 0.013: a mean loss of 0.0078, inside the margin, but a corrected interval over ten dependent scores admits 0.019, so this is an indicated reduction rather than demonstrated equivalence. Informed selection did *not* reliably beat a typical same-size subset—four of six sizes for permutation-based selection, two of six for forward—so on this redundant array how many sensors are used matters more than which. Degradation falls almost entirely on false alarms, rising from 4% to 30% while missed detections stay near zero. An SVM reproduces the curve's shape but meets the tolerance at $k = 5$, so the threshold is model-dependent where the shape is not. We further show that a channel carrying a label-correlated recording artefact is a learnable shortcut that marginal importance ranks last of eight, and give the diagnostic that exposes it. Code and stored predictions are available.**

***Index Terms*—fault diagnosis, rotating machinery, sensor selection, feature importance, machine learning, MaFaulDa**

I. Introduction

Instrumenting a rotating machine is not free. Every accelerometer carries a unit cost, a cable run, a mounting point that must survive vibration, an acquisition channel and a calibration interval. On a retrofit the mounting points are often the binding constraint: a casing never designed for instrumentation may offer two usable faces, not seven. The practical question is therefore not "what is the best achievable accuracy" but "how many sensors, and which, must I install."

The literature answers the first question far more often than the second. Much work proposes new representations and classifiers evaluated with the full sensor complement [1]. Work that does study reduction generally uses a heuristic search—greedy forward selection, a filter ranking, an embedded importance score [2]—and reports the subset the heuristic found without establishing where it sits among all subsets of its size. Whether the heuristic beat an arbitrary choice usually goes unmeasured.

We measure the degradation curve directly. With seven sensors an exhaustive sweep is tractable: $2^7 - 1 = 127$ subsets on every split is 1,270 model fits. Enumeration removes the search heuristic from the result. The distribution of all subsets of a size becomes the reference against which an informed choice can be judged, and the best subset of each size becomes an explicit upper bound—one reachable only by consulting test scores, which we therefore report as a bound and never as a result.

We fixed three questions in advance. **RQ1:** how does performance, overall and per class, change as the sensor count k falls from 7 to 1? **RQ2:** can a subset chosen from *training data only* match the full array? **RQ3:** which sensors contribute most, and does informed selection beat a typical subset of the same size? And three hypotheses, with thresholds fixed before any result was seen. **H1:** removing one sensor never raises mean macro-F1 by more than a comparability tolerance of 0.01 at any consecutive step. **H2:** some $k < 7$ reaches within 0.01 of the full array. **H3:** for each k from 1 to 6 the selected subset beats the median same-size subset (one-sided Wilcoxon [3], $n = 10$, $\alpha = 0.05$).

Our contributions are: an exhaustive rather than heuristic subset evaluation, giving the full distribution at every size and an explicit oracle bound; an evaluation holding out whole speed bands, with leakage prevented by assertions and its magnitude measured rather than assumed; selection confined to training data by a nested inner split; a demonstration that a label-correlated recording artefact is a learnable shortcut which marginal importance ranks last, with the procedure that exposes it; and a reproducible release of pipeline, predictions, every table and figure, and a 139-test suite. We also report two negative results: H3 fails, so on this array how many sensors are used matters considerably more than which ones; and a second model family reproduces the curve's shape but not the exact reduction threshold.

II. Related Work

A. Fault Diagnosis on Rotating Machinery

Vibration analysis is the established basis for diagnosing rotating-machinery faults. Accelerometers near the bearings capture the excitations produced by imbalance, misalignment and bearing defects, and conventional feature sets combine time-domain statistics with spectral descriptors such as band energies and dominant frequency [1], [4]. Accuracies above 98% have been reported on the dataset used here [5], which makes the evaluation protocol rather than the headline number the thing worth scrutinising.

B. Sensor and Feature Selection

Selection methods are conventionally grouped as filters, wrappers and embedded methods [2]; greedy forward selection is the archetypal wrapper and remains common in condition monitoring because it is cheap and its output is an interpretable ordering [4]. Its weakness is that it returns one subset and no context: without the distribution of all subsets of the same size there is no way to tell whether the selection was worth running. Enumerating every subset becomes intractable quickly—$2^n - 1$ grows past a billion by thirty sensors—but at seven it is 127, which is what makes this design possible.

C. Evaluation Methodology

Where a dataset is built by segmenting continuous recordings, the segments of one recording share operating conditions, mounting and noise realisation. Splitting them at random measures memorisation rather than generalisation. This is an active area with quantified effects, not an open gap. Hendriks *et al.* [6] show it for the CWRU bearing dataset, where the conventional split inflates reported accuracy, and build training and test sets from independent bearings instead. Wheat *et al.* [7] compare splitting methods across two further datasets and report accuracy differences exceeding 40 percentage points attributable to the split alone. The underlying failure mode is what Kaufman *et al.* [8] formalise as leakage, and the remedy is their learn–predict separation: keep every segment of a recording on one side of the split. We adopt that remedy rather than propose one; what we add is to *measure* the inflation on this dataset ($+0.0081$) and to go a level stricter by holding out whole operating speeds.

Second, permutation importance as Breiman introduced it [9] estimates a variable's *marginal* contribution. Strobl *et al.* [10] show this is biased under correlated predictors, since two variables carrying the same information can each score near zero when either substitutes for the other, so a near-zero score cannot distinguish a sensor carrying no information from one merely redundant. Their remedy is conditional permutation, which we do not adopt; we instead report each sensor's standalone score and its correlation to the others alongside the ranking (Fig. 2).

Positioning. Neither exhaustive enumeration nor leakage-aware evaluation is novel in itself: enumeration is a brute-force baseline that becomes available only because seven sensors is small, and the leakage literature above is well developed. The contribution is their combination and what it makes measurable. Reporting the *full distribution* of subsets at each size turns "is informed selection worth running?" into a testable question rather than an assumption, and it is what allows us to report a negative answer. Holding out whole operating speeds asks for cross-speed generalisation, which a split over recordings does not. Doing both, with selection confined to training data, is what lets the curve be read as a property of the sensors.

III. Dataset and Methodology

A. Dataset and Features

MaFaulDa [11] comprises 1,951 recordings from a laboratory rotating-machinery rig, each 5 s at 50 kHz, on a rig specified at 700–3600 rpm (about 12–60 Hz), with every condition recorded across that range. Eight channels are recorded: six accelerometer axes at the underhang and overhang bearings (`ub_ax`, `ub_rad`, `ub_tan`, `ob_ax`, `ob_rad`, `ob_tan`), a microphone (`mic`) and an analogue tachometer (`tach`). Which bearing carries the triaxial unit is not stated in the dataset documentation, and we do not infer it.

This study uses seven sensors. The tachometer is excluded because the channel carries a label-correlated recording artefact rather than a fault signature; the decision was taken before any classifier was fitted, on distributional evidence alone, and is examined in Section IV-G. Its features are still extracted so the comparison remains possible. Seven sensors give $2^7 - 1 = 127$ non-empty subsets.

The class label is the top-level folder, giving six classes: *normal*, *imbalance*, *horizontal* and *vertical misalignment*, *underhang* and *overhang*. **The two bearing classes name positions, not defects**—defect types (ball, cage, outer race) sit in subfolders and are merged into the two position classes, so a model here answers which bearing is faulty, not what kind of defect it has. We state this because the dataset's own web page words these classes differently.

Each recording is split into five equal non-overlapping windows of $N = 50{,}000$ samples (1.00 s), giving 9,755 windows. From each sensor in each window we compute **exactly 18 features**, identical in definition for every sensor, so the full array has $7 \times 18 = 126$. Holding the count identical is deliberate: a subset's score must reflect which sensors it contains, not how many columns they contribute.

Let x be the window after mean removal, $\bar{a} = \frac{1}{N}\sum |x_i|$, and $\varepsilon = 10^{-12}$ a guard in every denominator. Table I gives all 18.

Three conventions worth stating because they change the numbers. **Band energies and the spectral centroid are normalised** by total spectral energy T, so the seven bands are fractions summing to approximately one and are invariant to overall gain. **The spectrum is a power spectrum**, the squared magnitude of the real FFT, not an amplitude spectrum. And **`rms` uses the population form** $(1/N)$: standard deviation is excluded from the set precisely because, after mean removal, the population standard deviation ($\mathrm{ddof} = 0$) is *identically* `rms`. The sample form ($\mathrm{ddof} = 1$) differs by $\sqrt{N/(N-1)}$,

TABLE I

THE 18 FEATURES, COMPUTED PER SENSOR PER WINDOW ON THE MEAN-REMOVED SIGNAL x. THE SPECTRUM IS $S_j = |\mathrm{rfft}(x \cdot w_{\mathrm{Hann}})_j|^2$ WITH A SYMMETRIC HANN WINDOW, 25,001 BINS AND A RESOLUTION OF EXACTLY 1.00 HZ; $T = \sum_j S_j + \varepsilon$. THE SEVEN BAND EDGES (ℓ, h) ARE 5–250, 250–500, 500–1000, 1000–2000, 2000–5000, 5000–10000 AND 10000–25000 HZ.

Name	Definition	Unit
`rms`	$\sqrt{\frac{1}{N}\sum x_i^2}$	signal
`peak`	$\max \lvert x_i \rvert$	signal
`p2p`	$\max x_i - \min x_i$	signal
`crest`	$\mathtt{peak}/(\mathtt{rms} + \varepsilon)$	—
`kurtosis`	Fisher, $m_4/m_2^2 - 3$, biased	—
`skewness`	$m_3/m_2^{3/2}$, biased	—
`shape`	$\mathtt{rms}/(\bar{a} + \varepsilon)$	—
`impulse`	$\mathtt{peak}/(\bar{a} + \varepsilon)$	—
`clearance`	$\mathtt{peak}/\left(\frac{1}{N}\sum \sqrt{\lvert x_i \rvert}\right)^2$	—
`spec_centroid`	$\sum_j f_j S_j / T$	Hz
`dominant_freq`	$f_{\arg\max_{j \geq 1} S_j}$, DC bin excluded	Hz
`band_ℓ_h`	$\sum_{j \in [\ell, h)} S_j / T$, 7 bands	fraction

which at $N = 50{,}000$ is a factor of 1.00001 — numerically negligible, but the exclusion is justified by the exact identity, so the convention is stated rather than assumed. A test enforces that no two features have a constant ratio, using standard deviation against `rms` as a negative control.

Kurtosis and skewness are `scipy.stats.kurtosis` and `scipy.stats.skew` at their defaults: *biased* (population) moments $m_r = \frac{1}{N}\sum x_i^r$, Fisher's kurtosis, 0 for a Gaussian. The released environment pins SciPy 1.18.1, NumPy 2.5.3 and scikit-learn 1.9.1, because moment conventions are library-specific and a reimplementation using unbiased moments would not reproduce these columns.

Rotation frequency is taken from the file name, which the dataset uses to encode it, and is metadata rather than a model feature.

B. Cross-Validation and Leakage Control

The five windows of a recording are near-duplicates, so splitting them across the boundary measures memorisation. All results use `StratifiedGroupKFold`, 5 folds × 2 repeats, giving 10 paired scores; five folds alone cannot reach $p < 0.05$ in a one-sided Wilcoxon test, which is why the design repeats.

The primary protocol groups by rotation-speed band, ten quantile bands accepted by a guard requiring all six classes in every test fold. Holding out whole speed bands tests generalisation to unseen operating speeds, the realistic condition for variable-speed machinery. A secondary analysis groups by recording instead (Section IV-E). Both were run in full and both are reported, so no selection was made on which gave the more interesting result; the choice of primary was recorded before either experiment ran and rests on dynamic range—speed spans 0.112 macro-F1 between the best single sensor and all seven, against 0.020 for recording, leaving the curve room to move.

Isolation is enforced by assertions, not convention. A guard called by every selection and fitting path asserts no test index has entered training, and the split function asserts train/test and group disjointness on every split. A leak fails the run loudly.

The naive split is measured, not assumed. Splitting windows at random yields 0.9998 ± 0.0002 macro-F1 against 0.9917 ± 0.0066 grouped: an inflation of $+0.0081$ ($+0.0027$ under recording grouping). A random-window split on this dataset therefore reports a score that is perfect to three decimals and is not a measure of generalisation. It exists in our code only as this measurement. As sanity baselines the full array scores 0.1729 ± 0.0147 with shuffled labels and 0.0741 ± 0.0013 for a most-frequent dummy, against about 0.167 chance.

C. Scoring, Selection and Evaluation

Macro-F1 is primary. Detection errors are missed detection (fault windows predicted *normal*), false alarm (normal windows predicted as any fault) and misdiagnosis (fault windows predicted as a *different* fault).

The scoring label set is fixed once per outer fold, from the unique values of `y_true` in that fold's test set, and shared by every subset in it. The common default—the union of true and predicted labels—would let the denominator vary *between subsets inside one fold*, and every comparison here is paired within a fold.

Two selection strategies run, both confined to the training fold by an inner 75/25 split itself grouped by recording. *Grouped permutation importance* applies one shared row permutation to all 18 columns of a sensor together, so the sensor is destroyed as a unit; the five permutations are drawn once per split and reused for every sensor, pairing the drops. *Greedy forward selection* adds the sensor giving the highest inner-validation macro-F1 at each step, with deterministic ties.

One classifier is used for every subset, with no tuning and no feature scaling: a random forest of 200 trees with balanced class weights and a fixed seed. This is not a method paper; any difference between subsets must come from the sensors. All 127 subsets are evaluated on all 10 splits (1,270 score rows), predictions are stored, and every table and figure is computed from them without refitting. **Selected subsets are never refit**: with model and seed fixed, a selected subset's score *is* its entry in the exhaustive results and is looked up, so the two cannot disagree.

IV. RESULTS

A. RQ1: The Degradation Curve

Table II and Fig. 1 give the result. The full array reaches 0.9917 ± 0.0066. Averaged over all subsets of each size, performance falls monotonically—0.986 at $k = 6$, 0.969 at $k = 4$, 0.915 at $k = 2$, 0.797 at $k = 1$—while the spread widens from ± 0.005 at $k = 6$ to ± 0.077 at $k = 1$. The curve is strongly concave: the first three removals cost very little, the last two a great deal.

The widening spread is the practically important part. At $k = 1$ the choice of sensor moves the score across a range far larger than the loss from dropping seven sensors to four. A

TABLE II

MACRO-F1 VERSUS THE NUMBER OF SENSORS k. EACH SUBSET IS FIRST AVERAGED OVER THE 10 CROSS-VALIDATION SPLITS, THEN SUMMARISED ACROSS THE $\binom{7}{k}$ SUBSETS OF THAT SIZE. THE ORACLE COLUMN IS THE MAXIMUM OVER THOSE $\binom{7}{k}$ CANDIDATES CHOSEN USING TEST SCORES: IT IS AN UPPER BOUND, NOT AN ACHIEVABLE RESULT, AND ITS UPWARD BIAS GROWS WITH THE NUMBER OF CANDIDATES, PEAKING AT $k = 3$ AND $k = 4$. THE PERMUTATION- AND FORWARD-SELECTED COLUMNS ARE CHOSEN ON TRAINING DATA ONLY AND ARE THE HONEST COMPARISON. THE 10 SPLITS ARE NOT MUTUALLY INDEPENDENT.

k	$\binom{7}{k}$	All subsets	Oracle†	Perm.-sel.	Fwd.-sel.
1	7	0.797 ± 0.077	0.880	0.871 ± 0.058	0.882 ± 0.098
2	21	0.915 ± 0.037	0.967	0.948 ± 0.045	0.938 ± 0.066
3	35	0.953 ± 0.019	0.984	0.973 ± 0.038	0.949 ± 0.052
4	35	0.969 ± 0.012	0.988	0.984 ± 0.013	0.969 ± 0.027
5	21	0.979 ± 0.010	0.992	0.988 ± 0.012	0.986 ± 0.011
6	7	0.986 ± 0.005	0.991	0.989 ± 0.011	0.989 ± 0.010
7	1	0.992 ± 0.000	0.992	0.992 ± 0.007	0.992 ± 0.007

†Upper bound: selected using test scores.

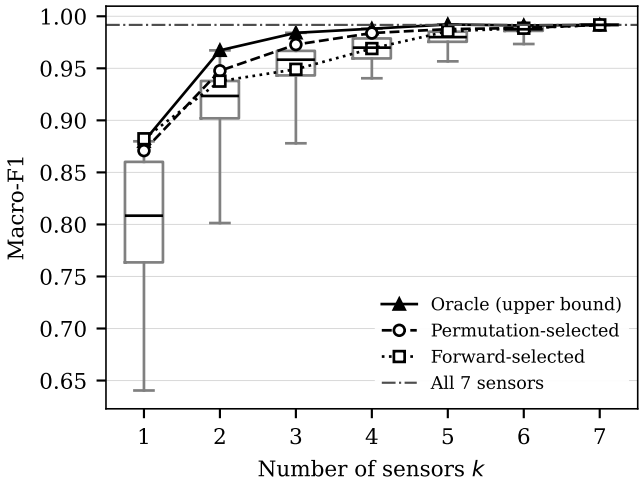


Fig. 1. Evaluation protocol: held-out speed bands (PRIMARY protocol). Macro-F1 against the number of sensors k. Each subset is first averaged over the 10 splits; the box plots then summarise those per-subset means across all $\binom{7}{k}$ subsets of each size. Whiskers span the full minimum to maximum range rather than 1.5×IQR: every subset is enumerated, so this is a complete population rather than a sample, and the extreme subsets are of direct interest. The oracle line is the maximum over those $\binom{7}{k}$ candidates selected using TEST scores and is an upper bound only. The 10 splits come from 2 repeats of 5-fold cross-validation over the same recordings and are therefore not mutually independent.

badly chosen small configuration is much worse than a small configuration as such.

H1 is SUPPORTED. On the permutation-selected curve (0.8709, 0.9476, 0.9728, 0.9839, 0.9877, 0.9887, 0.9917 for $k = 1 \ldots 7$) and the forward-selected curve (0.8823, 0.9377, 0.9490, 0.9690, 0.9856, 0.9885, 0.9917) there are **zero violations**: removing a sensor never raised the mean by more than 0.01. No sensor here is actively harmful.

The primary verdict rests on those two *fair* curves, whose subsets were chosen on training data. The oracle curve—0.8799, 0.9673, 0.9841, 0.9881, 0.9922, 0.9912, 0.9917—also shows no violation but is secondary and carries a caveat. The oracle at each k is a *maximum* over $\binom{7}{k}$ candidates (7, 21, 35, 35, 21, 7, 1), and a maximum over more candidates carries more upward bias, so the curve is inflated most at middle k independently of any sensor effect. An apparent violation there could be that bias rather than a harmful sensor, which is why the verdict does not rest on it; Table II prints the candidate count so the bias is visible.

TABLE III

NON-INFERIORITY OF THE PERMUTATION-SELECTED k-SENSOR SUBSET AGAINST THE FULL ARRAY. PAIRED DIFFERENCES ACROSS THE 10 SPLITS; INTERVAL AND TEST USE THE NADEAU–BENGIO CORRECTED STANDARD ERROR. THE SIX TESTS FORM ONE FAMILY, SO THE HOLM-ADJUSTED COLUMN IS THE ONE A CONFIRMATORY CLAIM MUST REST ON. "NOT SHOWN" MEANS THE EVIDENCE IS INSUFFICIENT, NOT THAT THE CONFIGURATION IS INFERIOR.

k	Mean diff.	95% CI	Raw p	Holm p	Verdict
1	+0.1208	[+0.0449, +0.1967]	0.9954	1.0000	not shown
2	+0.0441	[−0.0131, +0.1013]	0.8947	1.0000	not shown
3	+0.0189	[−0.0285, +0.0663]	0.6599	1.0000	not shown
4	+0.0078	[−0.0030, +0.0186]	0.3297	1.0000	not shown
5	+0.0041	[−0.0058, +0.0139]	0.1029	0.5147	not shown
6	+0.0030	[−0.0048, +0.0108]	0.0363	0.2178	not shown

B. RQ2: How Few Sensors Suffice

With the full array at 0.9917 and tolerance 0.01, the H2 threshold is 0.9817. **H2 is SUPPORTED at** $k = 4$: the permutation-selected four-sensor subset reaches 0.984 ± 0.013, while $k = 3$ reaches 0.973 and does not. The subset was chosen on training data alone, so this is a reduction a practitioner could actually have achieved. The oracle reaches the threshold at $k = 3$; that is an upper bound on what would be possible if the best subset of each size were known in advance, which it never is.

Where the tolerance comes from. The 0.01 margin was fixed before any result was seen, as the smallest macro-F1 difference this design can resolve: the full configuration's own spread across splits is ±0.0066, so a smaller margin would sit inside the measurement noise. It is a comparability threshold, not a claim about what an operator should tolerate, which depends on the cost of a false alarm against the cost of a channel.

A point estimate inside a margin is not demonstrated equivalence. H2 as specified compares means. Table III asks the inferential question instead, one-sided non-inferiority against the same 0.01 margin, with the standard error carrying the Nadeau–Bengio correction [12] because the paired differences are not independent.

No configuration demonstrates non-inferiority under this protocol. On raw p, only $k = 6$ would qualify. But the six tests are scanned precisely to find which k qualifies, which makes them one family; Holm adjustment takes the smallest to $0.0363 \times 6 = 0.2178$ and nothing survives. These six tests are therefore **exploratory**, and we present no k as a confirmed non-inferior configuration.

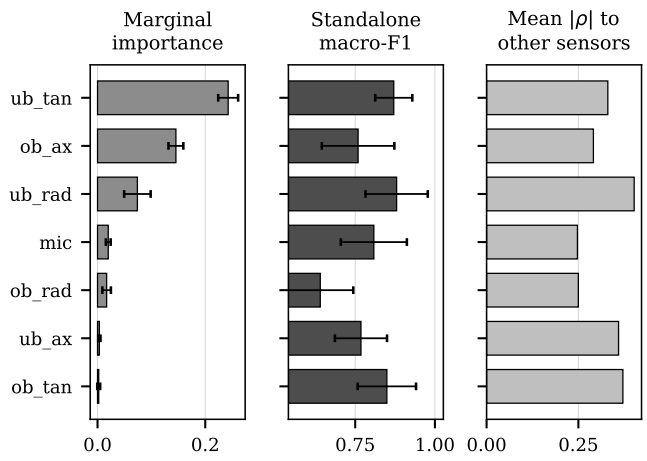


Fig. 2. Marginal importance, standalone performance and inter-sensor correlation for the seven channels, ordered by marginal permutation importance. LEFT: the drop in inner-validation macro-F1 when all 18 features of that sensor are permuted together, mean and standard deviation over the 10 splits; this is a MARGINAL contribution, measured with every other sensor present. CENTRE: the macro-F1 that sensor reaches on its own. RIGHT: its mean absolute Spearman correlation to the other six, over all 18 matched features. Reading the panels together separates the two cases a marginal measure cannot: `ob_tan` has the lowest marginal contribution of the set (0.0022) yet is the third best single sensor (0.8493), so its low score reflects redundancy rather than an absence of information. The 10 splits come from 2 repeats of 5-fold cross-validation over the same recordings and are therefore not mutually independent. Evaluation protocol: held-out speed bands (PRIMARY protocol).

At $k = 4$ the mean loss is $+0.0078$ but the interval reaches $+0.0186$, so a loss twice the margin is not excluded. With ten dependent scores the test has little power, so this does not settle equivalence either way rather than being evidence against it. H2 as pre-specified compares means and its verdict is unchanged by any of this.

The answer to RQ2. The smallest configuration whose *mean* falls within tolerance is four sensors, but "three sensors can be removed at no cost" overstates that: four is the indicated reduction and the interval in Table III is its risk.

C. RQ3: Which Sensors, and Does Selection Help

Fig. 2 gives three views of each sensor. The marginal ranking is dominated by `ub_tan` (0.2425 ± 0.0185) and `ob_ax` (0.1455 ± 0.0138), then `ub_rad` (0.0740 ± 0.0246); the other four have low marginal contribution. The ranking is stable—`ub_tan` appears in the permutation-selected subset in all 10 splits at every k, and `ob_ax` from $k = 2$ up.

The centre and right panels are why the ranking must not be read as information content. `ob_tan` has the lowest marginal contribution of all seven (0.0022 ± 0.0027) yet scores 0.8493 alone, third best of any single sensor, and has the second-highest mean correlation to the others (0.371). `ub_ax` behaves the same way (0.0034 marginal, 0.7680 standalone). In both cases the low marginal score reflects redundancy given the rest of the array, not a shortage of signal. Conversely `mic`, fourth of seven marginally (0.0199), has the *lowest* correlation to the rest (0.247) and enters the oracle subset at every size from $k = 2$ up: its information is unique rather than abundant. The most strongly coupled pair is `ub_rad`–`ub_tan`, correlating at $\rho = 0.888$ on RMS, and `ub_rad`—the best standalone sensor at 0.8799—enters the oracle subset at $k = 1$ then *drops out* at $k = 2$ and $k = 3$ once `ub_tan` is available.

H3 is NOT SUPPORTED. For permutation-based selection the chosen subset beat the median same-size subset significantly at four of six sizes ($k = 1$, $p = 0.0039$; $k = 2$, $p = 0.0039$; $k = 3$, $p = 0.0010$; $k = 5$, $p = 0.0156$) but not at $k = 4$ ($p = 0.0879$) or $k = 6$ ($p = 0.0742$); for forward selection at only two of six ($k = 1$, $k = 2$, both $p = 0.0010$). The pre-specified criterion required all six. Percentile ranks say the same more gently: the selected subsets sit between the 46th and 74th percentile of their size, so selection helps modestly and inconsistently rather than reliably finding a top subset.

Two post-hoc checks, neither of which disturbs the pre-specified verdict, both push in the same direction. Holm adjustment [13] for the family of six comparisons changes nothing material (four of six and two of six survive). More importantly, the ten scores are *not* mutually independent—two repeats reuse the same recordings, so training sets overlap heavily and the naive variance is too small—so we also report the Nadeau–Bengio corrected resampled t-test [12], which replaces the $1/n$ variance factor with $1/n+1/4$ for 5-fold. The correction is derived there for repeated random subsampling; the $1/(k-1)$ term is the usual adaptation to repeated k-fold. Under it only three of six survive for permutation selection and one of six for forward. H3 fails more decisively once dependence is accounted for, not less. We do not report a 5×2cv t-test [14]: that procedure requires five replications of *two*-fold cross-validation, whereas this design is two repeats of *five*-fold, and halving the training set would change the sensor-count curve itself rather than only its p-values.

D. Fault Sensitivity

Classes do not degrade together. At the full array every class is at or above 0.959 recall. As sensors are removed *normal* degrades first and furthest: 0.959 at $k = 7$, 0.922 at $k = 4$, 0.698 at $k = 1$. The two bearing classes are most robust, above 0.91 even at $k = 1$, with *imbalance* at 0.945; horizontal misalignment is the weakest fault class at 0.826.

In operational terms the three error types diverge sharply. Missed detection—a fault called healthy—is essentially flat and very low throughout, from 0.0001 at $k = 7$ to 0.0046 at $k = 1$: even one sensor almost never misses a fault outright. Misdiagnosis rises modestly, 0.0044 to 0.0892. What collapses is the false-alarm rate: 0.0408 at $k = 7$, 0.0776 at $k = 4$, 0.302 at $k = 1$. At one sensor roughly three in ten healthy windows are flagged faulty.

This asymmetry is the most useful finding in the per-class results. A reduced installation on this rig does not become dangerous—it does not start missing faults—it becomes *untrustworthy*, and loses the confidence of whoever answers its alarms. A single macro-F1 number conceals that.

E. Secondary Analysis: Held-Out Recordings

Repeating the analysis with recordings rather than speed bands as the grouping gives the same answers under a less demanding protocol. The full array reaches 0.9973 ± 0.0037; H1 is supported with zero violations on both fair curves; H2 is supported at $k = 2$, lower than the primary $k = 4$ as expected when the model is tested only at speeds it has seen; H3 is again not supported, four of six and two of six. The importance ranking is unchanged in order. Agreement on every verdict under a 0.020 dynamic range rather than 0.112 is the robustness evidence this analysis was run for. The non-inferiority picture does differ, informatively: under this easier protocol $k = 4$, 5 and 6 survive Holm adjustment across the same family of six tests, where under the primary protocol none does. Whether equivalence can be *shown* therefore depends on how demanding the evaluation is, not only on how many sensors are kept — which is itself a reason to be wary of reduction thresholds reported under recording-level splits.

F. Supplementary: A Second Model Family

Fixing one model isolates the sensor effect, but it leaves open whether the curve reflects the sensors or a tree-ensemble inductive bias. We therefore re-evaluated all 127 subsets on all 10 splits with an RBF support vector machine—a different family with a different bias—at scikit-learn defaults, with standardisation fitted inside the pipeline on the training fold only. The subsets compared at each k are those the primary analysis selected on training data; selection is not redone, so the question asked is whether the reduction the primary analysis found still holds under another model.

The shape replicates; the exact threshold does not. The SVM is uniformly weaker in absolute terms—0.9787 with all seven sensors against the forest's 0.9917—but the curve has the same concave form, the same monotone fall in the all-subsets mean (0.976 at $k = 6$ to 0.673 at $k = 1$), and the same sharp widening of spread at small k (± 0.005 to ± 0.116). **H1 is supported with zero violations**, as in the primary analysis. **H2, however, is met at $k = 5$ rather than $k = 4$**: the four-sensor subset reaches 0.9652 against a threshold of 0.9687 and falls just short.

The qualitative conclusions—graceful-then-sharp degradation, no actively harmful sensor, a substantial reduction available—are therefore not artefacts of the forest, while the precise number that can be dropped *is* model-dependent: a practitioner transferring the $k = 4$ figure to another classifier should re-derive it. Under the SVM the six-sensor subset marginally exceeds the full array (0.9822 against 0.9787), a gap well inside the 0.01 tolerance and consistent with the full array's larger variance under this model (± 0.026).

G. The Excluded Tachometer Channel

We ran the pipeline a third time with the tachometer included: eight sensors, 255 subsets, 2,550 score rows. We report it as a methodological contribution and a caution for other sensor-selection work, not as a result of this study.

TABLE IV
ACCURACY ON BEARING-CLASS WINDOWS, SPLIT BY WHETHER THE RECORDING CARRIES THE TACHOMETER ARTEFACT. A MODEL EXPLOITING THE ARTEFACT SHOULD DO MARKEDLY BETTER ON AFFECTED WINDOWS. COMPUTED FROM STORED PREDICTIONS OF THE 8-SENSOR RUN; 670 AFFECTED AND 10,040 UNAFFECTED WINDOWS.

Model	Affected	Unaffected	Difference
`tach` alone	0.8239	0.4068	+0.4171
`ob_rad` alone (control)	1.0000	0.8390	+0.1610
All 8 sensors	1.0000	0.9984	+0.0016
All 7 sensors	1.0000	0.9972	+0.0028

Why it was excluded. On 67 of the 1,951 recordings the tachometer trace separates from non-ball-fault windows at Cohen's d up to 6.58, and from *other* ball-fault recordings at d up to 4.54, while ball-fault recordings in general separate from the rest at only $|d| \leq 0.43$. A genuine fault signature would not separate a condition from other recordings of that same condition; this is a recording artefact. It is also label-correlated: all 67 carry *underhang* (59, 10.6% of that class) or *overhang* (8, 1.6%) labels and no other class is touched. Since RQ3 asks which sensors carry diagnostic information, a contaminated ranking would answer it falsely.

The exclusion changed no conclusion. The 8-sensor analysis gives identical H1 and H2 verdicts (fair $k = 4$, oracle $k = 3$), a full array of 0.9917 ± 0.0080, a permutation-selected curve matching to three decimals at every k, and the same oracle subset at every k—the tachometer never enters one. Had the exclusion produced a more favourable outcome, these would differ.

Marginal importance ranks the artefact last. Across the ten splits the tachometer is last of eight in eight splits and seventh in the other two, mean importance $+0.0002 \pm 0.0003$, two orders of magnitude below the next lowest (`ob_tan`, $+0.0023$). Alone it scores 0.2471 ± 0.0491, barely above the 0.1729 shuffled-label baseline and last of the eight. On this evidence the channel looks harmless.

It is not harmless. Marginal importance cannot say whether the artefact is *learnable*. Splitting the bearing-class windows into those from the 67 affected recordings and the rest answers it directly (Table IV). A model given only the tachometer classifies affected windows at 0.8239 against 0.4068 unaffected, a gap of $+0.4171$; a control sensor with no known artefact gaps by $+0.1610$, about a third as much. The artefact is learnable and a model with only that channel uses it. One qualification: the control's affected-window accuracy is at ceiling, so its gap is compressed from above.

Two things follow together. The exclusion was not merely precautionary—it removed a real shortcut. And with the other seven sensors present the gap disappears ($+0.0016$ and $+0.0028$, both at or near 1.0 on affected and unaffected alike), so the artefact did not distort the full-array results, which is why primary and supplementary analyses agree on every verdict.

The methodological point. Our own first reading was wrong. From *pooled* per-class recall the tachometer's elevated

bearing-class recall ($+0.221$) was indistinguishable from the control's ($+0.246$) — a correct observation from which we drew a wrong conclusion, because pooling averaged 670 affected windows with 10,040 unaffected and destroyed the contrast. Where a suspected artefact is confined to an identifiable subset, that subset must be isolated before any comparison is informative, and a near-zero marginal importance score is not evidence that a channel is safe to include.

V. Discussion

Instrumentation. On this rig and protocol, four well-chosen sensors reach a mean macro-F1 within the pre-specified comparability margin of the seven-sensor configuration, though as Section IV-B shows the available evidence does not establish non-inferiority. Read as a cost curve rather than an equivalence claim, the shape is still the actionable part: the decision is not a smooth cost/accuracy trade-off but a threshold. Above roughly four channels the mean gain from each additional sensor is smaller than the spread between splits; below it, cost rises steeply, two sensors to one costing more than seven to four. If the budget permits one, *which* becomes the dominant design decision—the spread across single sensors (± 0.077) is more than ten times that across six-sensor subsets (± 0.005).

Which faults go first. The class that degrades first is *normal*. Reducing sensors here does not produce a system that misses faults but one that cries wolf, false alarms rising from 4% to 30% while missed detection stays near zero: the risk of under-instrumenting is alarm fatigue, not undetected failure. Where a specific failure mode drives the business case the relevant curve is that mode's own recall—the bearing classes hold above 0.91 even at one sensor, so an installation targeting bearing faults could be reduced further than one also expected to certify a machine healthy.

Why these axes, physically. The ranking is not arbitrary with respect to the mechanics. The two dominant channels measure in different planes: a tangential axis in the plane of rotation and an axial axis along the shaft. Radial and tangential axes at a bearing both respond to once-per-revolution forcing from residual imbalance and to the housing's response to rolling-element impacts, which excite high-frequency structural resonances and are the basis of envelope diagnosis [15]. That shared forcing is the physical reason `ub_rad` and `ub_tan` are the most correlated pair in the array ($\rho = 0.888$) and why either largely substitutes for the other—high standalone scores, low marginal contribution. Axial response is excited by a different mechanism, the out-of-plane component that angular misalignment introduces, and two of the six classes here are misalignment conditions. An in-plane and an axial channel therefore sample complementary rather than overlapping physics, which is consistent with `ub_tan` and `ob_ax` ranking first and second and with the microphone—a non-contact channel subject to neither mounting path—being the least correlated of the seven. We offer this as an interpretation consistent with the measurements, not as something the experiment tested; the dataset does not document which bearing carries the triaxial unit, so we do not reason about specific mounting geometry.

Why selection helped so little. The failure of H3 and the correlation structure are one finding seen twice. Six of seven channels are accelerometers on the same rig measuring the same excitations through different mountings, so the array is substantially redundant. When many subsets of a size carry nearly the same information, most score nearly the same, the distribution is tight, and no selection method has room to distinguish itself—hence selected subsets in the 46th–74th percentile rather than the 90th. The practical reading is encouraging: on a redundant array a reasonable arbitrary choice of four sensors will usually do nearly as well as a carefully selected four. But it is a genuine negative result for selection methods, invisible without the exhaustive baseline, and it warns against pruning on marginal importance alone—which here would have kept redundant accelerometers and might have dropped the one genuinely complementary channel.

A. Limitations

Single dataset, and a rig rather than a plant. All results come from MaFaulDa, and the redundancy structure driving our H3 result is a property of this particular array. More importantly, the recordings come from a controlled rig running at steady speed under no applied load, in a laboratory acoustic environment, with faults introduced deliberately at known severities. Field machinery is not like that: load varies with process demand, speed transients and nearby equipment inject broadband noise, mounting degrades, and faults develop gradually rather than appearing at a set magnitude. Each of those raises within-class variance, and the absolute scores here—a full array at 0.9917 and even one sensor near 0.88—should be read as an upper envelope that benign conditions make possible, not as field performance. The *shape* of the degradation curve is the transferable claim; its height is not.

No cross-channel features. Each sensor contributes exactly 18 features computed from its own signal, which is what makes a subset's score attributable to the sensors rather than to a differing feature count. The cost is that quantities defined only *between* channels are absent: relative phase between orthogonal axes at a bearing, cross-spectra and coherence, and the orbit and full-spectrum representations that distinguish forward from backward whirl. These are standard in industrial diagnosis and are precisely the quantities a multi-sensor array is well placed to provide, so an array evaluated with them might degrade differently—plausibly more sharply, since such features vanish entirely when one of a pair is removed. Our curve therefore characterises sensor reduction under per-channel features, and extending it to cross-channel ones is open work.

Model dependence. A single fixed random forest carries the primary analysis, by design, so that differences between subsets are attributable to the sensors. Section IV-F tests how far that choice constrains the result: under an SVM the curve shape, H1 and the widening spread all replicate, but H2 is met at $k = 5$ rather than $k = 4$, so the *threshold* is model-

dependent even where the shape is not. Two families are not many, and neither consumes raw time series, where a model able to exploit temporal structure directly might weight the sensors differently.

Design-time reduction, not runtime loss. Every subset model is retrained on that subset. This answers which sensors to install, not what happens when an installed sensor fails mid-operation, which would require evaluating a full-sensor model under missing inputs.

Non-independent scores, and what $\pm$ means here. The ten scores come from two repeats over the same recordings, so they are not mutually independent. Every $\pm$ in this paper is a *standard deviation across the ten splits* — a description of how much the score moved between folds. It is *not* a confidence interval, and it is not an estimate of uncertainty in generalisation performance to new machines, which this design cannot measure at all from a single dataset. Where an inferential interval is meant, it is labelled as one and carries the Nadeau–Bengio correction (Table III). The H3 p-values are supporting evidence alongside effect sizes and percentile ranks, not standalone proof; the corrected test is itself an approximation to the dependence structure rather than a solution to it, so no strong significance claim rests on ten observations.

Tachometer channel quality. On the 67 affected recordings a rising-edge speed estimate from the tachometer overcounts, reporting up to 320.6 Hz against a true 50.79 Hz. Those traces swing about 22% wider (median standard deviation 2.118 against 1.742), lowering the midpoint threshold from about 1.97 to 0.80–1.27 and into the noise band. An FFT of the same channel recovers the correct frequency to within about 1.3 Hz, so the signal carries the information and only the threshold crossing fails. We therefore take rotation frequency from the file name. This is a property of the dataset's channel on those files rather than of the method, and it matters for any work deriving speed from it.

VI. Conclusion

We measured how fault-diagnosis performance depends on the number and choice of sensors by evaluating all 127 subsets of seven sensors with one fixed classifier, under cross-validation holding out whole rotation speeds, with every selection made on training data and three hypotheses fixed in advance.

Performance degrades gracefully then sharply: 0.9917 ± 0.0066 macro-F1 with seven sensors to 0.8799 with the best single sensor, with no step at which removing a sensor helped by more than 0.01 (H1 supported). **Four sensors chosen on training data reach 0.984 ± 0.013, a mean loss of 0.0078 against the full array** (H2 supported at $k = 4$; the test-selected oracle reaches it at $k = 3$, an upper bound and not an achievable result). Non-inferiority against the same margin is not demonstrated at *any* k once the six tests are adjusted for multiplicity, so this is an indicated reduction rather than demonstrated equivalence. Informed selection did not reliably beat a typical subset of the same size—four of six sizes for permutation-based selection, two of six for forward—so H3 is not supported, and on this redundant array how many sensors are used matters more than which. Degradation concentrates in false alarms, rising from 4% to 30% as the array falls to one sensor while missed detection stays near zero. Separately, a channel carrying a label-correlated recording artefact proved a learnable shortcut that marginal importance ranks last of eight, detectable only by isolating the affected recordings rather than examining pooled summaries.

What is demonstrated, and what is not. Demonstrated: within this dataset, pipeline, classifier and protocol, four sensors carry a mean macro-F1 loss of 0.0078 against seven, and the curve is concave with no step at which a sensor is actively harmful. Not demonstrated: that the two configurations are statistically equivalent; that four sensors suffice in service; or that the threshold holds for other classifiers, which the SVM already contradicts. This is a method for determining a reduction curve, not licence to remove three channels.

Future work should test whether a four-sensor threshold persists in service; whether the redundancy that defeats selection here is a property of this array or of accelerometer arrays generally; and how the curve changes for a model consuming raw time series. Runtime sensor loss—how a full-sensor model behaves when an installed sensor fails—is a distinct experiment this design does not address. The pipeline, stored predictions, every table and figure, and the 139-test suite that verifies them are available at https://github.com/Narenkaarthik/sensor-reduction-study, with `requirements-lock.txt` pinning the versions that reproduce the numbers. The MaFaulDa archive is obtained separately from [11].

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